

CSCI 160 Homework 1

1. Add the following binary numbers: $10101 + 11011 + 111$.
2. Express each of the following numbers in decimal:
 - (a) $(11010.101)_2$
 - (b) $(4021.2)_5$
 - (c) $(127.4)_8$
 - (d) $(B65F)_{16}$
3. Convert decimal 247 to binary.
4. Convert decimal 153 to octal.
5. Multiply the binary numbers 10110×101 .
6. Perform the following conversions:
 - (a) Convert $(10110011.1011)_2$ to octal and to hexadecimal.
 - (b) Convert $(4F2A)_{16}$ to binary.
 - (c) Convert $(572.6)_8$ to binary.
7. Find the following complements:
 - (a) the 1's and 2's complement of 10110100
 - (b) the 9's and 10's complement of 246700
 - (c) the 16's complement of $C3DF$
8. Perform each subtraction using the indicated complement method:
 - (a) using 2's complement: $1010100 - 1000011$
 - (b) using 10's complement: $72532 - 3250$
 - (c) using 1's complement: $1100100 - 1010011$
9. Add the unsigned decimal numbers $791 + 658$ in BCD.
10. Convert decimal $+29$ and $+49$ to the signed-2's-complement representation using eight bits, then compute $(+29) + (-49)$ and convert the result back to decimal.
11. Binary logic. Let x , y , and z be binary variables.

- (a) Write the truth tables for the three basic logic operations: AND ($x \cdot y$), OR ($x + y$), and NOT (x').
- (b) Draw the three corresponding logic gates for the previous operations. Label your output as z and mark your inputs.